ODE-FFN: Progress Report 2

Independent Study — Deep Learning

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4 November 2026

1 Project and progress

This project studies FFNs designed around the continuous dynamics of flow matching, with the goal of improving the blocks inside JiT. Starting from two Taylor maps around SiLU, we explored local clocks, the generator P, polynomial order, nonlinear response and learned directions. The clearest recent finding is that a bounded ISRU response improves vector-field fitting. Sampling gains are smaller and depend on the ordinary model used for comparison.

Synthetic tests isolate the FFN in a width-64, three-block, time-conditioned MLP. Independent noise–data pairs train velocity from $z_t = tX + (1-t)e$ on analytic mixtures. Oracle MSE measures fitting; sliced Wasserstein distance (SW2) measures generation. Earlier studies use 5k updates; recent studies use 20k, batch 512, and Heun sampling. Budgets and solver settings accompany each comparison.

2 Local dynamics: clocks, order and optimization

The original scalar map at $K = 2$ is

$$T(h) = h + s(h)Ph + \tfrac{1}{2}s(h)^2P^2h, \qquad s(h) = 0.2 + 0.8\,\sigma(g(h)/10). \tag{1}$$

Clocks and order. The input clock differs from FM time and contributes to the full Jacobian. On Ring8, input gates reduce samples beyond 0.6 from every mode center: 19.9% to 8.3%, versus SwiGLU's 3.4% (parameter counts differ). Freeing fixed-clock effective matrices improves fitting. Exact frozen solves, state feedback, clock reuse, biases and augmented states did not improve the quality–cost tradeoff across tasks. Scalar K1 improves all eight matched K1/K2 field comparisons. At 80k on Ring8, raw K1 retains a fitting gain against matched SwiGLU64, while SwiGLU wins on sampling and with larger widths.

Optimization. Muon approximately orthogonalizes FFN momentum updates; other parameters retain AdamW. We measured both updates and learned P.

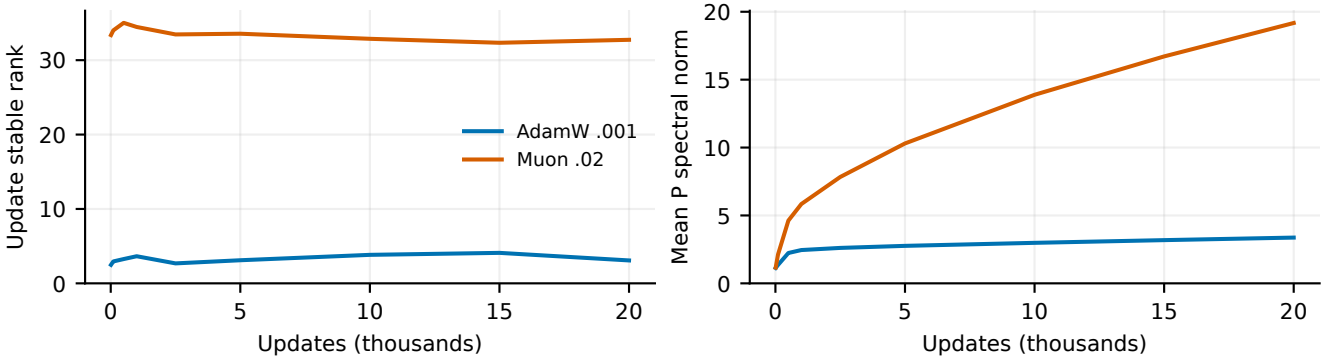


Figure 1: Product16 scalar K2, six-matrix means. Matrix rates: AdamW .001, Muon .02. Update stable rank is $\|\Delta P\|_F^2/\|\Delta P\|_2^2$. Muon broadens the update spectrum while P's spectral norm grows.

At 20k, AdamW/Muon update stable ranks are 3.08/32.74 and mean P norms 3.37/19.17. Field MSE improves from 0.03343 to 0.02385; SwiGLU192 reaches 0.01567. Norm-matched momentum does not reproduce all gains. Decay and training-time flattening harm fitting; flattening also worsens SW2.

3 Changing the complete field and its directions

Full-gradient readout. We tested whether structure in the complete predicted field matters more than a constraint on hidden P. For a vector backbone G, predict

$$v(x,t) = \nabla_x[\tfrac{1}{2}x^\top G(x,t)] = \tfrac{1}{2}[G(x,t) + J_G(x,t)^\top x]. \tag{2}$$

This matches the gradient-field structure of the analytic mixture velocities. At 5k, original-ODE Product16 MSE falls from 0.325527 to 0.037320; ordinary gradient SwiGLU reaches 0.033826 at nearly equal dense inference work. Both backbones benefit from the readout. Derivative-path and local-adjoint controls show that localizing the operation loses much of the improvement. On the Gaussian task, field fitting improves while sampling worsens.

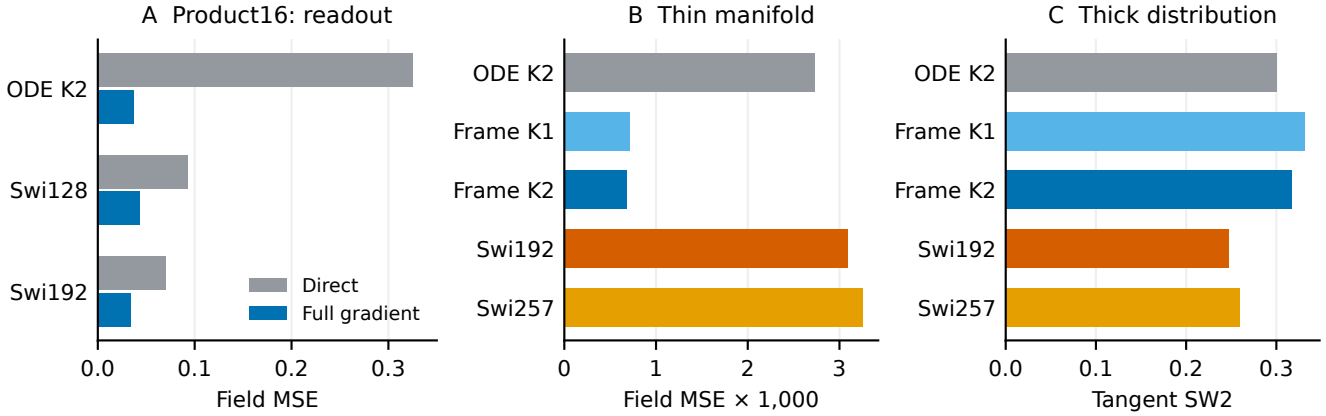


Figure 2: Lower is better. A: full-gradient readout helps both ODE and ordinary backbones. B: conditional frames improve thin-target field fitting. C: their sampling advantage fails on the thick target. Each panel has its own metric and axis scale.

Input-dependent directions. To change the directions in which P acts without changing its pointwise spectra, we tested $Q(h)T_K(s(h)P)Q(h)^\top h$ with an orthogonal learned frame. On a 2D mode plane embedded in 128D, frame K2 reaches MSE 0.000685 and K1 reaches 0.000713, versus SwiGLU192's 0.003095. K1 already has better tangent SW2 than K2. Adding variance outside the plane reverses the sampling result: frame K2 gives 0.3173 versus SwiGLU192's 0.2471. Follow-ups varying the quadratic coefficient, action and gradient route retain this task dependence. Direction adaptation is most useful when the target has a thin mode plane.

Nonlinear internal flow. We tested whether linear dynamics were too restrictive by splitting $dy/dr = Py - y^{\odot 3}$ into linear steps and the exact response $\phi_s(u) = u/\sqrt{1+2su^2}$. Product16 field fitting improves, but freeing the two matrices around that response works better:

5k Product16	Original K2	Tied cubic	Free ISRU	SwiGLU160
Field MSE	0.325527	0.092938	0.037159	0.078178

Table 1: Same earlier protocol. Free ISRU is an ordinary independent-matrix control, with a different parameterization and preserved special initializer.

Its advantage reverses on dependent-sign and half-mixture laws. Noisy-label, affine-head, frozen-feature and local-teacher studies leave part of the fitting gap unresolved. The strong Product16 response result motivated the bounded input-feature experiments below.

4 Input-dependent features and current evidence

The diagonal construction uses $D(h) = \mathrm{diag}(d(h))$ and $A(h) = PD(h)$. At K1,

$$F_1(h) = h + P[d(h) \odot h], \qquad J_{F_1} = I + P[\mathrm{diag}(d) + \mathrm{diag}(h)J_d]. \tag{3}$$

Controller tests cross rank-32 linear/MLP gates with sigmoid/affine outputs. Direct features (constant, h, sigmoid, $h+1$, $|h|$) isolate function and sign; radial controls isolate amplitude. Raw h helps Ring8 fitting, while absolute and radial variants do not improve the overall tradeoff.

To bound features, ISRU-D sets $d_i = (1+1.2h_i^2)^{-1/2}$, so $q_i = h_i/\sqrt{1+1.2h_i^2}$ and $q_i' = (1+1.2h_i^2)^{-3/2}$, retaining two K1 maps around SiLU. Free ISRU instead uses two independent matrices and a different initializer.

Ring8 FFN	Opt.	Field MSE	SW2, banks	SW2, pool
ISRU-D	A	**0.00977**	0.17110 (0.04721)	0.13261
Free ISRU	A	0.01000	0.17930 (0.04728)	0.14467
SwiGLU64	A	0.01372	0.19453 (0.06503)	0.17771
SwiGLU64	M	0.01128	0.17145 (0.05310)	0.13643
SwiGLU192	M	0.01123	0.18399 (0.05724)	0.15769

Table 2: 20k, Heun128. A: AdamW .0003; M: selected Muon .02. Banks: mean (SD) over eight common 4,096-sample banks. Pool: follow-up combining 32,768 saved samples. ISRU-D/free: 56,706 parameters, 55,616 dense MACs; Swi64: 56,766 and 55,232. MACs exclude pointwise work.

ISRU-D fits better than all four controls at each checkpoint; sampling is close to selected Swi64. From 15k to 20k, its MSE improves from 0.01026 to 0.00977 while SW2 rises from 0.16054 to 0.17110. Every ordinary control also samples worse at 20k.

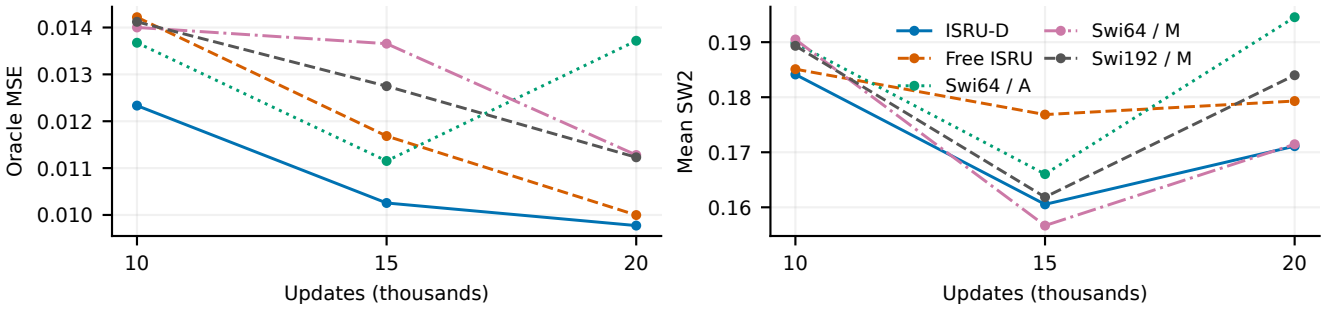


Figure 3: Ring8 checkpoint comparison. Left: oracle MSE. Right: mean SW2 of eight common banks, Heun128. Better fitting does not preserve sampling order.

On Product16 (20k, Heun32), MSE is 0.01606 for ISRU-D and 0.01536 for free ISRU; SW2 is 0.08574 for ISRU-D and 0.07921 for SwiGLU192. Its expanded sampling check is pending. Muon worsens both ISRU fits at the tested rate.

5 Hypotheses for the next two weeks

1. **Feature response explains optimizer effects.** On the same minibatch, compare norm-matched updates, changes in Pq, full-field loss and P spectra. Use their predictive value to choose the quantity to regulate.
2. **Error location explains the fitting–sampling mismatch.** At common transport states, compare ISRU-D, free ISRU and selected Swi64 against the analytic field by time, direction and full-field Jacobian. Localized late errors would motivate a targeted dynamics change; a shared response pattern would favor feature-function development.